Green World Multi-Role Platform

System Specification

1. Overview

Green World is a multi-role distribution, e-commerce, appointment booking, and delivery management platform designed for distributors, customers, store owners, drivers, and platform administrators.

The platform enables:

- Product distribution and online sales
- Distributor store management
- Multi-store operations
- Driver delivery tracking and earnings management
- Appointment booking for health scans and wellness checks
- Head office reporting and analytics
- Subscription-based store activation

The ecosystem includes:

- Super Admin Panel
- Distributor Panel
- Customer Storefront
- Driver Mobile App
- Appointment Booking Module
- Reporting & Analytics Module

The backend is built using Node.js, Express.js, MongoDB, and Firebase Authentication.

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2. High-Level Architecture

Backend

- Node.js
- Express.js
- MongoDB with Mongoose
- Firebase Admin SDK
- Redis (optional for queues and caching)

Frontend Applications

Super Admin Panel

- HTML
- CSS
- Vanilla JavaScript

Distributor Panel

- HTML
- CSS
- Vanilla JavaScript

Customer Panel

- HTML
- CSS
- Vanilla JavaScript

Driver Mobile App

- React Native
- Expo
- TypeScript

2.1 Roles

The platform supports the following roles:

Role	Description
Super Admin	Platform owner and system administrator
Store Owner	Distributor with an active store
Distributor	Distributor buyer using a distributor code
Customer	Regular retail customer
Driver	Delivery driver

Each role is authenticated through Firebase Authentication and authorized using role-based middleware.

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3. Registration & Authentication

3.1 Registration Types

Users must select one of the following account types during registration:

- Customer
- Distributor

Distributor Registration Requirements

Distributor registration requires:

- Valid Green World distributor code
- Email address
- Password
- Phone number
- Profile information

The system stores:

```
- distributor_code
- user role
- verification status
```

3.2 Distributor Verification System

The platform supports distributor verification using unique codes issued by Green World head office.

Distributor Code Features

- Unique distributor code per distributor
- Distributor code stored on user profile
- Used for reporting and analytics
- Used to validate distributor registration

Optional Admin Verification System

Admin can:

- Upload CSV/Excel lists of valid distributor codes
- Automatically verify distributor accounts
- Reject invalid distributor registrations

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4. Multi-Store System

Distributors can open and manage their own stores within the platform.

Store Features

- Multiple stores per distributor
- Unique public store page URL
- Store-level inventory management
- Store-specific pricing
- Store analytics and reporting
- Store delivery management

Example Store URL

```
/store/{store_slug}
```

Store Owner Capabilities

Store owners can:

- Manage products
- Manage inventory
- Manage customer orders
- View analytics
- Manage appointments
- Track delivery operations
- Track driver payouts

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5. Store Subscription System

Yearly Store Activation Fee

Stores require a yearly subscription fee for activation.

Subscription Rules

- Subscription valid for 12 months
- Store becomes inactive after expiry
- Expired stores cannot accept orders
- Renewal required to reactivate store

Subscription Features

- Subscription expiry tracking
- Renewal reminders
- Payment tracking
- Store lock/unlock system

Optional:

- Email reminders
- Push notifications
- Auto-renew support

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6. Project Structure

6.1 Root Structure

```
backend/                # Node.js + Express backend
super-admin-panel/      # Admin panel
store-owner-panel/      # Store/distributor panel
customer-panel/         # Customer storefront
driver-app/             # Driver mobile app
```

6.2 Backend Structure

```
backend/
  src/
    config/
      database.js
      firebase.js
      env.js
      redis.js

    models/
      User.js
      Distributor.js
      Store.js
      Product.js
      StoreProduct.js
      Order.js
      Driver.js
```

```
    DriverEarning.js
    Address.js
    Payment.js
    Appointment.js
    Subscription.js
    DistributorCode.js
    Report.js

  controllers/
    authController.js
    superAdminController.js
    distributorController.js
    storeController.js
    productController.js
    orderController.js
    customerController.js
    driverController.js
    appointmentController.js
    subscriptionController.js
    reportingController.js

  routes/
    authRoutes.js
    superAdminRoutes.js
    distributorRoutes.js
    storeRoutes.js
    productRoutes.js
    orderRoutes.js
    customerRoutes.js
    driverRoutes.js
    appointmentRoutes.js
    subscriptionRoutes.js
    reportingRoutes.js

  middleware/
    authMiddleware.js
    roleMiddleware.js
    validationMiddleware.js

  utils/
    emailService.js
    notificationService.js
    reportGenerator.js
    helpers.js

  app.js
```

```
  server.js
  package.json
```

6.3 Super Admin Panel

```
super-admin-panel/
  index.html

  css/
    style.css
    dashboard.css
    responsive.css

  js/
    app.js
    distributors.js
    products.js
    orders.js
    drivers.js
    reports.js
    subscriptions.js

  pages/
    distributors.html
    products.html
    stores.html
    orders.html
    drivers.html
    users.html
    appointments.html
    reports.html
    subscriptions.html
    settings.html
```

Admin Features

- Manage distributors
- Upload distributor codes
- Manage products
- Manage stores
- Manage subscriptions
- View reports
- Manage appointments

- Track driver payouts
- Export reports

6.4 Distributor / Store Owner Panel

```
store-owner-panel/
  index.html

  css/
    style.css
    dashboard.css

  js/
    app.js
    stores.js
    products.js
    orders.js
    appointments.js
    reports.js
    subscriptions.js

  pages/
    stores.html
    store-products.html
    orders.html
    inventory.html
    appointments.html
    reports.html
    profile.html
```

Distributor Features

- Create and manage stores
- Manage inventory
- View reports
- Manage appointments
- Track earnings
- Track subscriptions

6.5 Customer Panel

```
customer-panel/
  index.html

  css/
    style.css
    shop.css
    cart.css

  js/
    app.js
    shop.js
    cart.js
    checkout.js
    orders.js
    appointments.js

  pages/
    shop.html
    product-detail.html
    cart.html
    checkout.html
    orders.html
    order-tracking.html
    appointments.html
    booking-history.html
    addresses.html
    profile.html
```

Customer Features

- Browse products
- Place orders
- Book appointments
- Track orders
- Manage addresses
- View appointment history

6.6 Driver App

```
driver-app/
  app/
    (tabs)/
      _layout.tsx
      index.tsx
      orders.tsx
      active-delivery.tsx
      earnings.tsx
      profile.tsx

    login.tsx
    register.tsx

  components/
    OrderCard.tsx
    DeliveryStatus.tsx
    EarningsCard.tsx
    MapView.tsx

  services/
    api.ts
    auth.service.ts
    order.service.ts
    driver.service.ts
    earnings.service.ts
```

Driver Features

- Accept deliveries
- Manage delivery status
- View earnings
- Track payouts
- View completed deliveries
- Navigate using maps

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7. Database Design

7.1 Core Collections

users

Fields:

```
firebaseUid
email
role
name
phone
distributor_code
verificationStatus
```

---

distributors

```
userId
companyName
contactNumber
status
```

---

distributorCodes

```
code
isUsed
assignedTo
createdAt
```

---

stores

```
distributorId
name
slug
description
logo
```

```
subscriptionStatus
subscriptionExpiry
isActive
```

---

subscriptions

```
storeId
startDate
expiryDate
status
paymentReference
```

---

products

```
name
sku
description
basePrice
category
imageUrl
```

---

storeProducts

```
storeId
productId
storePrice
stock
isActive
```

---

orders

```
customerId
buyerType
storeId
distributorCode
items
subtotal
```

```
deliveryFee
total
paymentStatus
orderStatus
driverId
addressId
```

---

drivers

```
userId
vehicleType
vehicleNumber
licenseNumber
status
currentLocation
```

---

driverEarnings

```
driverId
orderId
deliveryFee
driverAmount
platformAmount
payoutStatus
```

---

appointments

```
customerId
storeId
serviceType
appointmentDate
appointmentTime
status
paymentStatus
```

---

reports

```
reportType
startDate
endDate
generatedBy
fileUrl
```

---

8. Authentication & Authorization

Authentication Flow

1. User registers/login via Firebase Authentication.
2. Firebase returns JWT token.
3. Client includes:

```
Authorization: Bearer <token>
```

1. Backend verifies token using Firebase Admin SDK.
2. Backend attaches decoded user to request.

Role-Based Access Control

Middleware

```
authMiddleware.js
roleMiddleware.js
```

Route Examples

```
requireRole('superAdmin')
requireRole('storeOwner')
requireRole('distributor')
requireRole('customer')
requireRole('driver')
```

---

9. Orders & Checkout Logic

Orders must store:

- Buyer type
- Distributor code
- Store ID
- Payment status
- Order status
- Driver assignment

Buyer Types

```
customer
distributor
```

This enables:

- Head office reporting
- Distributor purchase tracking
- Store analytics
- Sales reporting

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10. Delivery System & Driver Earnings

Delivery Fee Split

Delivery fees are automatically split:

Recipient	Percentage
Driver	75%
Store / Platform	25%

Driver Dashboard Features

Drivers can:

- View assigned deliveries
- Accept deliveries

- Track earnings
- View payout status
- View completed deliveries

Admin & Store Owner Features

Admins and store owners can:

- View driver earnings
- Track payout history
- Mark payouts as paid
- Generate payout reports

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11. Appointment Booking Module

The system includes a health scan and wellness appointment booking module.

Customer Features

Customers can:

- Select service type
- Select store
- Select appointment date
- Select appointment time slot
- View booking history
- Pay online (optional)

Store Owner Features

Store owners can:

- View appointments
- Confirm appointments
- Cancel appointments
- Complete appointments
- Manage schedules

Appointment Statuses

```
pending
confirmed
cancelled
completed
```

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12. Reporting Module

The platform includes exportable reporting tools for head office.

Reporting Features

- Distributor purchase reports
- Store sales reports
- Driver earnings reports
- Subscription reports
- Appointment reports

Export Formats

Supported formats:

- Excel (.xlsx)
- CSV
- PDF

Excel and CSV are recommended.

Filters

Reports can be filtered by:

- Date range
- Distributor code
- Store
- Driver
- Status

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13. Technology Stack

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose
- Firebase Admin SDK

Frontend Panels

- HTML
- CSS
- Vanilla JavaScript

Mobile App

- React Native
- Expo
- TypeScript
- React Navigation / Expo Router
- React Native Maps

Client-Side API & State

- Axios / Fetch
- Zustand

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14. Implementation Phases

Phase 1 – Backend Core

- Express setup
- MongoDB connection
- Firebase integration
- Core models

Phase 2 – Authentication & Roles

- Firebase authentication
- Role middleware
- Distributor code verification

Phase 3 – Super Admin Panel

- Distributor management
- Product management
- Reporting module
- Subscription management

Phase 4 – Distributor & Customer Panels

- Store management
- Inventory
- Orders
- Checkout
- Appointment booking

Phase 5 – Driver App

- Delivery tracking
- Driver earnings
- Payout system
- Maps integration

Phase 6 – Reporting & Analytics

- Exportable reports
- Dashboard analytics
- Head office reporting

Phase 7 – Testing & Deployment

- End-to-end testing
- Production deployment
- Monitoring & backups

15. Scalability Considerations

The system is designed for scalability using:

- Modular backend architecture
- Firebase authentication
- MongoDB scalability
- Redis caching
- Background workers
- Real-time notifications
- Mobile-first architecture

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16. Future Enhancements

Potential future improvements:

- Loyalty rewards
- Referral system
- SMS notifications
- AI-powered analytics
- Live driver tracking
- Multi-language support
- Advanced subscriptions
- Multi-region support
- POS integration
- Accounting integrations

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17. License

This project is proprietary software intended for internal or commercial business use.